VICOBA Backend API Specification

Tech Stack: FastAPI (Python), PostgreSQL, SQLAlchemy (ORM), Alembic (Migrations), Pydantic (Data Validation).

1. Core Backend Features

- **Async Performance:** Utilizing FastAPI's native async/await for high-throughput transactional processing, especially when waiting for third-party Mobile Money APIs.
- **Role-Based Access Control (RBAC):** Distinction between standard Members, Group Admins (Chairperson, Treasurer, Secretary), and Super Admins.
- **ACID Compliant Transactions:** Ensuring that share purchases and loan disbursements are strictly isolated and consistent.
- **Automated Swagger UI:** Built-in API documentation via /docs (FastAPI default).
- **Maker-Checker Logic:** Financial actions (like loan approval or cycle closure) require multi-admin approval.
- **Background Tasks:** Using FastAPI BackgroundTasks or Celery for non-blocking operations like sending SMS notifications.

2. Global Error Handling Strategy

FastAPI handles payload validation automatically (returning 422 Unprocessable Entity). For business logic, the system will use standard HTTP status codes:

- 400 Bad Request: Business rule violation (e.g., "Loan amount exceeds 3x shares").
- 401 Unauthorized: Missing or invalid JWT token.
- 403 Forbidden: Valid token, but insufficient permissions (e.g., Member trying to close a cycle).
- 404 Not Found: Resource does not exist.
- 409 Conflict: Concurrency issues (e.g., "Transaction already processed", "Phone number already registered").
- 500 Internal Server Error: Unexpected backend failure.

3. API Endpoints Specification

A. Authentication, Registration & User Profile

1. Register User (Self-Registration or Claim Shadow Profile)

- **Endpoint:** POST /api/v1/auth/register
- **Behavior:** Registers a new user. If a "shadow profile" already exists (because an Admin added this phone number to a group ledger previously), it updates the profile status to active and links the device. Includes basic KYC (National ID).
- **Payload:**

```
{
  "phone_number": "+255700000000",
  "full_name": "Amina Juma",
  "national_id": "19900101-12345-00001-2",
  "pin": "1234"
}
```

- **Success (201 Created):** { "message": "User registered, OTP sent for verification", "reference_id": "uuid" }
- **Failure (409 Conflict):** If the phone number is already actively registered.

2. Request OTP / Login

- **Endpoint:** POST /api/v1/auth/request-otp
- **Behavior:** Checks if the user exists and is active. If yes, generates a 4-digit OTP, triggers the SMS gateway, and returns a reference ID.
- **Payload:** { "phone_number": "+255700000000" }
- **Success (200 OK):** { "message": "OTP sent successfully", "reference_id": "uuid" }
- **Failure (404 Not Found):** If the user is not registered.

3. Verify OTP & Get Token

- **Endpoint:** POST /api/v1/auth/verify
- **Behavior:** Validates the OTP against the reference ID. Returns a JWT access token.
- **Payload:** { "reference_id": "uuid", "otp": "5678" }
- **Success (200 OK):** { "access_token": "eyJhbG...", "token_type": "bearer" }

4. Get Member Passbook (My Profile)

- **Endpoint:** GET /api/v1/users/me
- **Behavior:** Returns the authenticated user's digital passbook, including KYC status, active groups, total shares, active loans, and recent transactions.
- **Response (200 OK):**

```
{
  "user_id": 45,
  "name": "Amina Juma",
  "kyc_verified": true,
  "groups": [
    {
      "group_id": 101,
      "group_name": "Mavuno Women Group",
      "role": "member",
      "total_shares": 15,
      "share_value_total": 75000,
      "social_fund_balance": 10000,
      "active_loan": { "amount_due": 50000, "next_payment_date":
"2026-09-14" }
    }
  ]
}
```

B. Group (Chama) Management

1. Create a VICOBA Group

- **Endpoint:** POST /api/v1/groups
- **Behavior:** Creates a new group, sets bylaws, and initializes the first cycle. The user creating the group becomes the Chairperson automatically.
- **Payload:**

```
{
  "name": "Mavuno Women Group",
  "share_value": 5000,
  "max_shares_per_meeting": 5,
  "interest_rate_monthly": 0.10,
  "meeting_frequency": "weekly"
}
```

- **Success (201 Created):** { "group_id": 101, "status": "active" }

2. Add Member to Group (Invite / Shadow Profile)

- **Endpoint:** POST /api/v1/groups/{group_id}/members
- **Behavior:** Admin adds a user to the group. If the phone number is not registered in the system, it creates a "shadow profile" so they can still be tracked on the ledger. Sends an SMS invite to download the app.
- **Payload:** { "phone_number": "+255700000001", "role": "member", "full_name": "David" }
- **Success (201 Created):** { "member_id": 46, "status": "added_as_shadow", "sms_sent": true }

3. Get Group Ledger / Member List

- **Endpoint:** GET /api/v1/groups/{group_id}/members
- **Behavior:** Returns all members and their current standing (shares, social fund, loans). Used by Admins to view the master ledger.

C. Meeting & Attendance Management

VICOBA operations revolve around meetings. Transactions should ideally be linked to a specific meeting session.

1. Start a Meeting

- **Endpoint:** POST /api/v1/groups/{group_id}/meetings
- **Behavior:** Opens a new meeting ledger.
- **Payload:** { "scheduled_date": "2026-09-07" }
- **Success (201 Created):** { "meeting_id": 12, "status": "in_progress" }

2. Record Attendance & Fines

- **Endpoint:** POST /api/v1/groups/{group_id}/meetings/{meeting_id}/attendance
- **Behavior:** Admins mark attendance. Automatically generates fines for late or absent members based on group bylaws.
- **Payload:**

```
{
  "attendees": [45, 46, 47],
  "late": [{"member_id": 48, "fine_amount": 1000}],
  "absent": [49]
}
```

D. Fund Mobilization (Deposits)

1. Record Deposit (Shares & Social Fund)

- **Endpoint:** POST /api/v1/groups/{group_id}/transactions/deposit
- **Behavior:** Initiates a deposit for shares and social fund. Triggers Mobile Money USSD Push to the member's phone.
- **Payload:**

```
{
  "member_id": 45,
  "meeting_id": 12,
  "shares_bought": 3,
  "social_fund_amount": 2000,
  "payment_method": "mobile_money"
}
```

- **Success (202 Accepted):** { "transaction_id": "tx-8989", "status": "pending_payment" }
- **Failure:** 400 Bad Request (Shares exceed max meeting limit).

E. Credit Allocation (Loans & Repayments)

1. Apply for a Loan

- **Endpoint:** POST /api/v1/groups/{group_id}/loans/apply
- **Behavior:** Validates eligibility (e.g., requested amount <= shares * 3).
- **Payload:** { "amount": 150000, "purpose": "Agriculture", "duration_months": 3 }
- **Success (201 Created):** { "loan_id": 202, "status": "pending_approval" }

2. Approve & Disburse Loan

- **Endpoint:** POST /api/v1/groups/{group_id}/loans/{loan_id}/approve
- **Behavior:** Admin approves the loan. Triggers B2C Mobile Money payout to the member.
- **Payload:** { "admin_pin": "1234" }
- **Success (200 OK):** { "status": "disbursed", "payout_reference": "MPESA-9923" }

3. Repay Loan

- **Endpoint:** POST /api/v1/groups/{group_id}/loans/{loan_id}/repay
- **Behavior:** Initiates a USSD push for a member to pay back their loan installment.
- **Payload:** { "amount": 55000 }
- **Success (202 Accepted):** { "transaction_id": "tx-9001", "status": "pending_payment" }

F. Social Fund (Insurance Claims)

1. Request Social Fund Payout

- **Endpoint:** POST /api/v1/groups/{group_id}/social-fund/claim
- **Behavior:** Member requests emergency funds (e.g., medical, bereavement).
- **Payload:** { "amount": 50000, "reason": "Medical Emergency" }
- **Success (201 Created):** { "claim_id": 5, "status": "pending_group_approval" }

G. Cycle Closure (Share-out)

1. Calculate & Execute Cycle Share-out

- **Endpoint:** POST /api/v1/groups/{group_id}/cycles/{cycle_id}/close
- **Behavior:** Locks the cycle. Sums all interest and fines. Calculates dividends proportionally based on shares held.
- **Payload:** { "admin_signatures": [4, 7, 12], "confirmation_pin": "9999" }
- **Success (200 OK):**

```
{
  "status": "closed",
  "total_profit_distributed": 1200000,
  "dividend_per_share": 2400,
  "member_payouts_initiated": true
}
```

H. Webhooks (Third-Party Integrations)

1. Payment Gateway Callback

- **Endpoint:** POST /api/v1/webhooks/payments
- **Behavior:** Webhook for Selcom/ClickPesa. Updates transaction status (for shares, fines, or loan repayments) and sends SMS receipt.
- **Payload (From Provider):**

```
{
  "transaction_reference": "tx-8989",
  "status": "SUCCESS",
  "amount": 17000
}
```

- **Success (200 OK):** { "acknowledged": true }